14.3 A 28nm 17.83-to-62.84TFLOPS/W Broadcast-Alignment Floating-Point CIM Macro with Non-Two's-Complement MAC for CNNs and Transformers

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The rapid advancement of artificial-intelligence (AI) models has increased demand for high-precision and energy-efficient edge-AI chips. Floating-point (FP) support is essential for high-precision neural-network (NN) training and inference; yet FP incurs higher energy and area overhead due to complex FP multiplication and accumulation (MAC) operations. Digital compute-in-memory (DCIM) and floating-point CIM (FP-CIM) [1-10] have emerged as promising techniques to improve energy efficiency with higher accuracy. Previous FP-CIM implementations [1-7] achieved good performance through various alignment schemes and computing processes. However, as illustrated in Figure 14.3.1, the implementation of a digital-domain FP-CIM faces several challenges: (1) the difficulty of balancing FP-computation precision and input reusability, as alignment operations are unfriendly to CIM structure; (2) a large performance loss or area overhead due to peripheral parallel-alignment schemes; and (3) huge digital-MAC dynamic-energy consumption due to low 2's-complement (2C) negative-weight sparsity, coupled with an additional sign-bit computation overhead in digital CIM. This work presents a hierarchical broadcast-alignment non-2's-complement-MAC (B-A-N2CMAC) FP-CIM macro, featuring (1) a broadcast input and embedded lightweight convertor structure to enable BF16/INT8 MAC operations with an improved input reusability; (2) an embedded area-efficient adaptive-alignment scheme with a dual-bit serial MAC; and (3) a format-mixed N2CMAC to reduce dynamic circuit activity and signed computation overhead. A 28nm 64kb B-A-N2CMAC FP-CIM macro is fabricated to support FP-MAC operations using BF16 and INT8 representations. This CIM macro achieved an energy efficiency of 62.84TFLOPS/W for BF16 and 90.15TOPS/W for INT8.

Figure 14.3.2 illustrates the overall structure of the proposed B-A-N2CMAC FP-CIM macro, which includes 16 triple-stacked arrays (TS-As), a hierarchical input buffer, a calibrated accumulator and quantizer (CA&Q), and peripheral circuits. Each TS-A computes one output channel: a TS-A comprises of (1) an embedded serial-input converting unit (ESICU), (2) a 16×256 6T SRAM array, and (3) a format-mixed non-2's-complement MAC unit (N2CMACU). Unlike previous FP-CIM designs [2,3], global floating-point or integer inputs (G_{IN}), from the hierarchical input buffer, can be reused by 16 TS-As to maximize the array-utilization rate. Each G_{IN} consists of an 8b parallel-exponent input (E_{IN}) and a 2b serial-mantissa input (M_{IN}). ESICU, which contains an align-signal generator (ASG) and 16-lightweight converters (LCs), is designed to achieve a local 2b-serial alignment for G_{IN} depending on the exponent data read from the SRAM array. This approach eliminates the need for large-area subtractors and barrel shifters, compared to typical digital-alignment methods [1,2,5]. To fully utilize small-absolute-value negative-weight sparsity, N2CMAC computation is proposed to enable weight-mapping using a sign-magnitude data format; where, a signed floating-point or integer MAC is divided into an unsigned MAC and a sign-compensation, performed in the TS-A and CA&Q. The CIM macro supports four modes: one conventional read and write mode, and 3-computation modes: BF16A, BF16B, and INT8. BF16A and BF16B modes differ based on the preserved bit-width of the aligned mantissa, 10b for BF16A and 8b for BF16B, which are sufficient to handle the target CNN and transformer networks.

Figure 14.3.3 illustrates the ESICU structure and the embedded adaptive 2b-serial alignment scheme. The ESICU is composed of an ASG and 16 LCs: the ASG consists of 16 exponent adders (EAs), a maximum-value finder (MVF), a shift-destination tracer (SDT), and 16 equality comparators (ECs); the LC consists of an adaptive-reset register chain (ARRC), a self-alignment controller (SAC), and an output select and invert unit (OSIU). After the generation of exponent sums by EAs, the SDT value is determined based on the maximum sum from MVF, and decreases by 1 each cycle. Each EC then compares the SDT value with the higher 8 bits of its corresponding exponent sum (SUM[8:1]) and generates shift control signals (SHIFT0 - 15) for the LCs. If SHIFT = 0, the ARRC is in the store state, sequentially storing the serial minimum values; else, SHIFT = 1, the ARRC is in the shift state, sequentially outputting the previously stored minimum values: shifting out 2b per clock cycle. Then, OSIU generates OUT[1:0] and stores it as local input L_{IN}[1:0] for N2CMAC operations. The proposed adaptive 2b-serial alignment scheme reduces area overhead by 36.23%, compared to a traditional subtractor and barrel-shifter alignment schemes [1-6].

Figure 14.3.4 illustrates the detailed implementation of the format-mixed N2CMAC computation, along with the memory-bank configuration and waveforms for the 3 computation modes. Unlike previous 2's-complement MAC operations [11-13], weights are stored in the TS-As using a sign-magnitude format. The format-mixed N2CMAC computes MAC results for 2's-complement inputs and sign-magnitude weights, resulting in a 2's complement output. Computation consists of 4 steps: (1) the hierarchical input buffer provides G_{IN}, including its 8b parallel exponent in BF16A/B modes and a 2b serial M_{IN} in BF16A/B and INT8 modes; (2) the LC serially converts G_{IN} to L_{IN}, performing serial alignment in BF16A/B modes, and input-sign invert in BF16A/B and INT8 modes ({~L_{IN}[m-1], L_{IN}[m-2:0]}); (3) the digital multiply unit (DMU) inverts L_{IN} based on the sign of mantissa weight (M_W), multiplies L_{IN} by the magnitude of M_W, and the products are accumulated in an adder tree (AT); (4) the CA&Q shifts and accumulates the AT output over multiple cycles to generate PMACV, which is calibrated by adding a pre-set compensation value to the accumulator to obtain a MACV result: 20b result for INT8, 23b for BF16A, 21b for BF16B. The proposed format-mixed N2CMAC computation divides a signed MAC operation into an unsigned MAC and a sign-compensation. The unsigned MAC eliminates the overhead of sign-bit computation in the DMU and AT, while the sign-compensation value can be obtained either through two additional cycles or pre-set offline. This value is then shared across multiple cycles, further optimizing efficiency. In this work, memory-bank configuration is achieved through the MUX shared by adjacent columns. In BF16A/B modes, a staggered-mapping scheme is used for exponents and mantissa, while in INT8 mode the column-select (CS) signal functions as a column decoder, increasing the storage-compute ratio (SCR).

Figure 14.3.5 highlights the proposed schemes' performance. The serial-alignment and serial-MAC schemes proposed in this work achieve a 3.83× and 1.56× reduction in area and a 1.71× and 1.20× reduction in power, compared to prior parallel-alignment schemes using a parallel MAC [6] and a parallel-alignment scheme using a serial MAC [1,2]. The proposed N2CMAC computation reduces the required number of computing cycles for different networks with varying input precisions: a 19.80 and 16.50% reduction for 8b and 10b inputs using ViT (DeiT-S) @ImageNet. The use of sign-magnitude formatted weights significantly improves the sparsity of negative INT8 weights: showing a 1.83× bit-level sparsity increase for ResNet50 compared to a 2's complement format. The format-mixed N2CMAC circuit achieves a 1.96, 1.25, and 1.18× lower power consumption, latency, and area overhead compared to typical 2's complement MAC circuits [1,2].

Figure 14.3.6 presents measured results for the test chip, fabricated using a 28nm CMOS technology, implementing a 64kb B-A-N2CMAC FP-CIM macro. For BF16A/B-mode MAC operations, the measured access time is 5.4ns at 0.9V for BF16A/B inputs and weights and FP32 outputs. The maximum energy and area efficiency are 62.84TFLOPS/W and 697.17GFLOPS/mm² for BF16A/B MAC operations from a 0.55 - 0.9V supply when running ResNet50 @ImageNet. Compared to prior SRAM CIMs [1-5], this work improves the IN × W × memory density × norm. energy efficiency × output ratio FoM by 1.93 - 116.90×. This work achieves a 79.818% inference accuracy when used on ViT @ImageNet. Figure 14.3.7 shows the die photograph and chip performance summary table. All input/weight sparsity and toggle rates are bit-level and all simulation and measurements are obtained at room temperature (300K).

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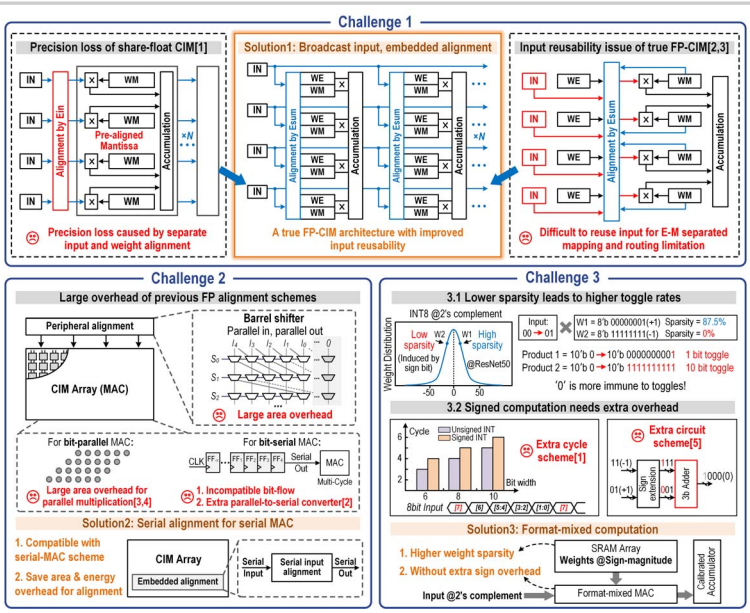


Figure 14.3.1: FP-CIM design challenges and proposed solutions.

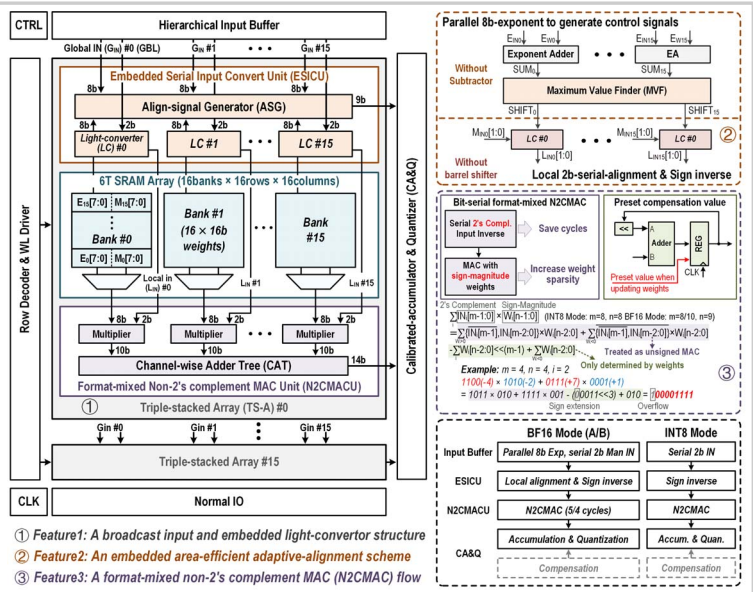


Figure 14.3.2: Overall structure with key features and BF16/INT8 CIM reconfiguration of the proposed B-A-N2CMAC FP-CIM macro.

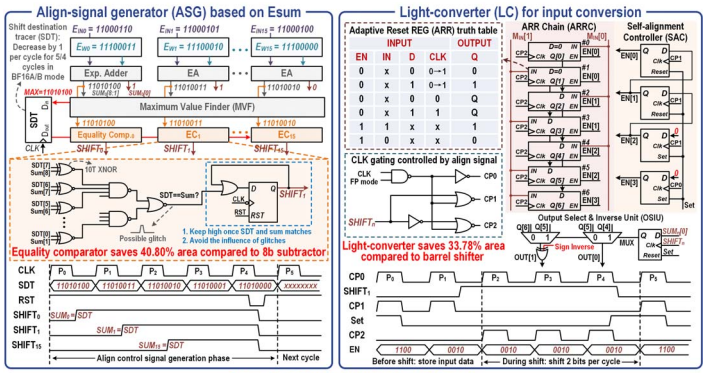


Figure 14.3.3: The ESICU structure for the embedded 2b-serial alignment, ASG for control signal generation, LC for input conversion (top). An example of ESICU workflow in BF16 mode (bottom).

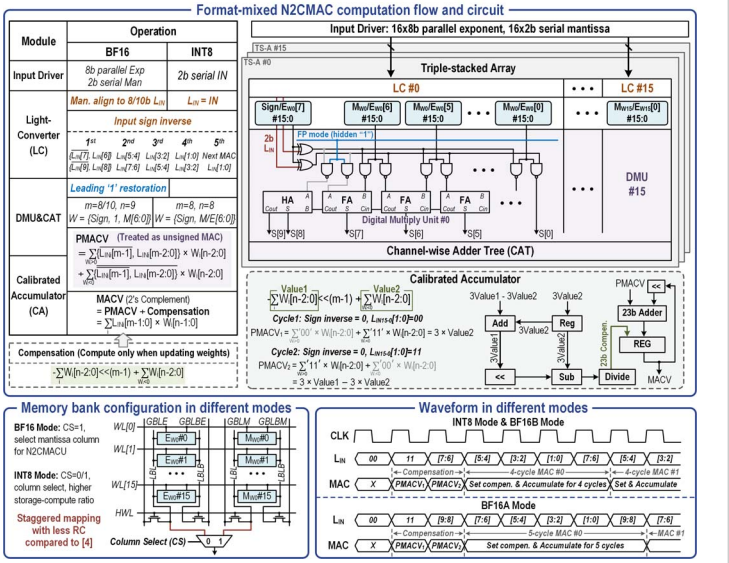
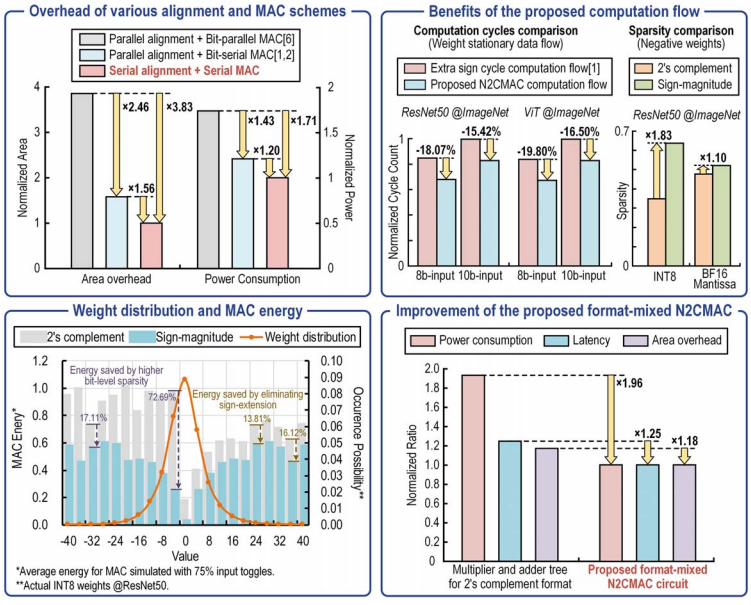


Figure 14.3.4: Detailed implementation of format-mixed N2CMAC computation (top). Memory bank configuration and MAC operation waveforms for 3 computation modes (bottom).



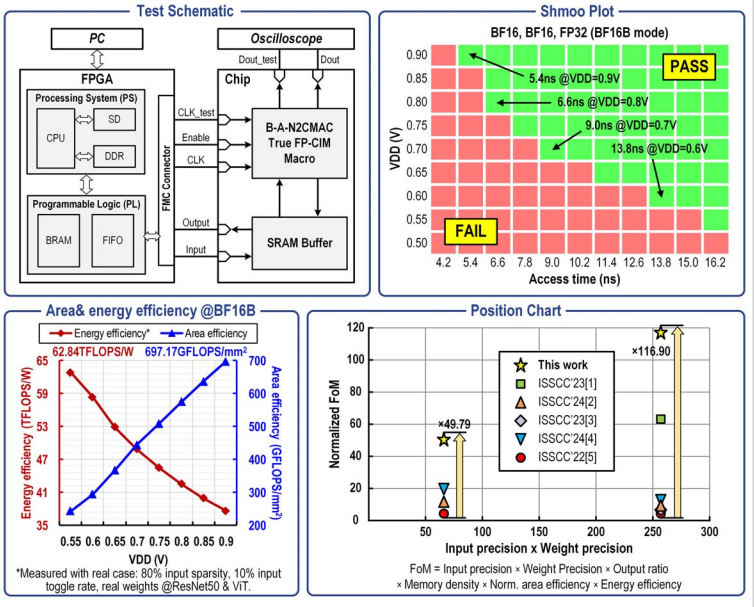
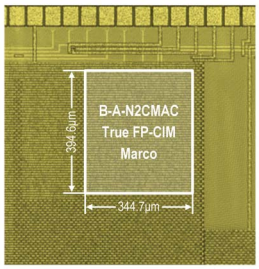


Figure 14.3.5: Proposed CIM macro simulated performance.

Figure 14.3.6: Measurement results and FoM comparison to prior work.

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[1]Measured with worst case under 0.9V. Worst case: 25% input sparsity, 50% input toggle rate, 15% weight sparsity.
[2]Measured with real case under 0.55V. Real case: 80% input sparsity, 10% input toggle rate, real weights @ResNet50 & ViT.
[3]Software baseline is 76.140%.
[4]Data-efficient Image Transformer-Small (DeiT-S), software baseline is 79.834%.
[5]Software baseline is 57.974%.

CHIP SUMMARY		
Technology	28nm CMOS	
Cell structure	6T+ESICU&N2CMAC	
Macro size	64Kb	
Macro area (mm^2)	0.136	
Input precision (bit)	8	BF16
Weight precision (bit)	8	BF16
Output precision (bit)	20	FP32
Supply voltage (V)	0.55-0.9	
Output ratio	1	
Access time (ns)	4.8	5.4/6.8
Memory Density (Kb/mm^2)	470.59	
Energy efficiency (TOPS/W) (TFLOPS/W)	[1]25.47-[2]90.15	[1]17.83-[2]62.84
Inference accuracy (%) ([3]ResNet50 @ImageNet)	76.030	76.130/76.130
Inference accuracy (%) ([4]ViT @ImageNet)	-	79.806/79.818
Inference mAP50 (%) ([5]YOLOv8 @COCO)	56.968	57.970/57.973

Figure 14.3.7: Die micrograph and chip performance and key-metric summary table.

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